HASNAE ELBALLAOUI

AI & Digital Transformation Engineering Student

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Seeking a final-year internship (PFE) in Artificial Intelligence starting January 2027

PROFILE

AI engineering student (ENSAH) who carried out a PFA in NLP and information retrieval at 3D Smart Factory: a CV–job matching pipeline evaluated on 344 CVs. Experience complemented by projects in machine learning, RAG and data engineering. Proficient in Python, SQL and Java; analytical rigor, autonomy and a taste for concrete projects.

EDUCATION

École Nationale des Sciences Appliquées d'Al Hoceïma (ENSAH) *2022 – Present*

Engineering Degree – Artificial Intelligence and Digital Transformation (preceded by an integrated preparatory cycle)

Main courses: Cloud Computing, Machine Learning, Deep Learning, NLP, Cybersecurity, DevOps, Data Engineering, Software Engineering

PROFESSIONAL EXPERIENCE

PFA Intern – NLP / AI | 3D Smart Factory (Mohammedia, remote) *Jul. 2026 – Sept. 2026*

Deep, personalized hybrid pipeline for candidate–job matching

- CV–job matching pipeline: Sentence-Transformers embeddings, FAISS index, Cross-Encoder re-ranking, skills-oriented score and personalization.
- Evaluation on 344 CVs and 28 offers: NDCG@5 of 0.900 vs 0.849 (BM25) and 0.790 (random); diagnosis and correction of an initially counter-productive Cross-Encoder.
- Streamlit application, job collection via APIs (Remotive, France Travail), partial Airflow/Docker orchestration.

Web Developer Intern | ALKHEBRA PRO *Aug. 2025 – Sept. 2025*

- Design and development of the company's complete website (HTML, CSS, JavaScript), responsive interface.
- Full cycle handled: structure, interface, implementation and testing.

ACADEMIC PROJECTS

Software Engineering Chatbot – NLP Comparative Study (GitHub) *Python, PyTorch, FLAN-T5, LoRA, FAISS*

- Comparison of three architectures: Seq2Seq with attention, fine-tuned FLAN-T5 (LoRA) and RAG.
- RAG (FAISS, BM25, Cross-Encoder) selected: more effective with fewer hallucinations; evaluated with BLEU, ROUGE, MRR.

Job Intelligent – Job-offer Data Pipeline (GitHub) *Python, Airflow, MinIO, PostgreSQL, Docker*

- ETL pipeline unifying 4 job sources (Kaggle, France Travail, Rekrute, Adzuna), orchestrated with Airflow and containerized with Docker.
- MinIO data lake (Bronze, Silver, Archive, Gold) and PostgreSQL star-schema data warehouse, Power BI reporting.

CAN 2025 – Winner Prediction (GitHub) *Python, Scikit-Learn, Random Forest*

- Merged 5 sources into a base of 8,297 matches; 20 engineered features (FIFA ranking, form, fatigue, head-to-head).
- Tuned Random Forest (minimal overfitting), then 1,000 Monte Carlo simulations of the full tournament.

ATTENDIFY – Face-recognition Attendance (GitHub) *Python, OpenCV, LBPH, Tkinter, MySQL*

- Desktop app: face detection (Haar Cascade) and recognition (LBPH), student management, MySQL storage and CSV export.

GestionPFE – Final-year Project Defense Scheduling (GitHub) *Java, Spring MVC, Apache POI*

- Web app: Excel import and constraint-based schedule generation (max 4 defenses per professor per day, 1-hour break, unavailability), validation and Excel/Word exports.

Conversational IoT Assistant – Cloud Deployment (GitHub) *Python, Bot Framework, Azure, GitHub Actions*

- Chatbot controlling simulated devices, deployed to Azure Web App with a GitHub Actions CI/CD pipeline.

SKILLS

Programming languages: Python, Java, SQL, PHP, HTML, CSS, JavaScript
AI & Machine Learning: Machine Learning, Deep Learning, NLP, LLMs & RAG, Computer Vision, Scikit-Learn, PyTorch, Pandas, NumPy, OpenCV, Sentence-Transformers, FAISS, Cross-Encoders, BM25
Tools & environment: Linux, Git/GitHub, Docker, Airflow, Hadoop, Spark, MinIO, Flask, Streamlit, Odoo, PostgreSQL, MySQL, Power BI, pipelines ETL, Agile (Scrum)
Soft skills: Curiosity, autonomy, analytical mind, teamwork, problem solving, adaptability, attention to detail

LANGUAGES

Arabic: native | French: fluent | English: fluent